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# Crystallization dynamics and interfacial structure during growth of thin Nb films on Cu by high-power impulse magnetron sputtering: A molecular dynamics study

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ABSTRACT

Improving the performance of niobium (Nb)-coated copper (Cu) Superconducting Radio Frequency (SRF) cavities is crucial for next-generation particle accelerators. Yet, performance is often limited by the reduction of the cavities' superconducting properties at high accelerating fields, a behavior linked to Nb thin-film growth defects. Using large-scale molecular dynamics (MD) simulations, we systematically investigate the influence of high-power impulse magnetron sputtering (HiPIMS) deposition parameters, specifically ion energy and fraction of ionized species in the deposition flux, on the growth of Nb films on Cu(1 1 1) substrates. We find that high ion energies and ionization fractions induce significant surface roughening during growth, which is in agreement with the available experimental data. Moreover, our simulations reveal that this behavior is driven by the formation and rapid crystallization of a metastable amorphous Nb-Cu phase at the interface. Critically, we discover a nonmonotonic relationship between the composition of the amorphous layer and the concentration of residual vacancy defects. The defect density is minimized at an optimal composition of approximately 45% Cu, increasing for the interfaces with a lesser or greater amount of Cu compared to Nb. This reveals that the optimization of SRF cavities requires not only an increase in ion energy but also an accurate and careful tuning of deposition parameters to engineer the ideal interfacial composition, thereby minimizing the number density of performance-degrading defects.

## 1. Introduction

Advances in charged-particle beam accelerators are crucial for several disciplines, such as medicine, biology, and solid-state physics [1–3]. Superconducting radio-frequency (SRF) cavities improve accelerator performance, offering significant advantages, including low heat dissipation losses on the cavity wall [4,5]. Bulk SRF cavities made of niobium (Nb) exhibit excellent performance due to their low surface resistance [6], but they are characterized by high manufacturing costs. Nb-coated copper (Cu) cavities have therefore emerged as a promising cost-effective alternative that combines good thermal conductivity with potential material savings [7,8]. A persistent limitation for Nb-on-Cu cavities is the Q-slope: an accelerating-field-dependent reduction of the cavity quality factor (Q) [9]. One major contributor to Q-slope has been suggested to be the formation of nanoscale porosity and related defects during thin-film growth, which can locally degrade device superconducting behavior [9–12].

Nb-coated SRF Cu cavities are commonly fabricated by magnetron sputtering [13,14]. This thin-film deposition method exhibits pronounced line-of-sight characteristics and, as such, yields non-uniform film morphology, thickness, and interface structure when used for growing films on complex-shaped substrates [15–18], including elliptical (U-shaped) SRF Cu substrates [19]. In a previous combined experimental and simulation work [19], we showed that direct current magnetron sputtering (DCMS) yields non-uniform, porous, columnar morphologies on surfaces obliquely exposed to the flux (vs. uniform columnar film morphologies for surfaces facing the deposition flux). However, high-power impulse magnetron sputtering (HiPIMS) tends to produce more uniform, denser films across all substrate orientations. Moreover,

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through simulations, we identified three key structure-forming mechanisms responsible for the overall morphology and microstructure of films grown by HiPIMS: (i) deflection of ionized species towards substrate-normal directions; (ii) bombardment-induced interruption of island growth; and (iii) ballistic atomic rearrangement coupled with local dynamic thermal annealing.

Although comparison of film morphologies grown with different methods favors HiPIMS, several experimental studies have focused on the effect of the substrate bias electrostatic potential (voltage) on the microstructure of HiPIMS-grown films [20,21] to further improve their quality. The substrate bias voltage ($V_b$) is a key parameter to adjust the energy of the (positively) charged species that arrive at the substrate during magnetron-sputter deposition [22] and to increase metal ion capture due to deflection of ionized fluxes towards the normal substrate surface [23,24]. An example of this effect can be seen in the study that covers a wide window of substrate biases from −25 V to −125 V [25]. Here *Wang et al.* utilized local third-harmonic probe ($P_{3f}$) measurements, which are sensitive to the vortex nucleation and qualitatively correlate with cavity Q factor, to demonstrate that films deposited at an intermediate bias of −75 V exhibit the most effective suppression of surface defects and the best RF response. Similarly, Carlos et al. reported that while increasing the bias from −50 V improves the critical current density ($J_c$), the performance peaks at −75 V and subsequently declines as the bias is further increased to −125 V [26]. Both studies confirm that while moderate ion energy is beneficial, excessive bias leads to a degradation of device performance [25–27].

Despite this empirical optimization evidence, the atomistic mechanisms underpinning the effect of the bias voltage on the microstructural features (including defects) that govern the performance of Nb-coated Cu SRF cavities remain unclear.

Molecular dynamics (MD) is a well-established tool for studying energetic film deposition with atomistic resolution and precision. Previous MD studies of Nb film growth under conditions consistent with HiPIMS deposition have shown that the ion energy and the ion-to-neutral ratio of the deposition flux strongly influence the mobility, composition, and structure at the film-substrate interface, as well as defect generation [19,28–30]. However, the transient atomistic mechanisms governing the amorphization of the Nb–Cu interface under HiPIMS conditions, and the relationship between interfacial composition and the resulting vacancy defect density, remain poorly understood. Moreover, MD work on other film-substrate systems (e.g., Cu on Ni trenches; Cu seed layers for interconnect metallization) has demonstrated that incident energy and thermal conditions can substantially alter surface roughness, interfacial mixing, and subsequent recrystallization behavior [31,32]. Importantly, MD simulations provide access to atomistic processes occurring on sub-nanosecond timescales at the growth front, phenomena that are extremely challenging to observe experimentally due to their transient nature and the spatial resolution required. While chamber pressure and mean free path effects are relevant for understanding flux distributions in the plasma phase, our MD approach focuses specifically on the interfacial processes at the substrate surface, where particle energies and fluxes are prescribed based on values typical for HiPIMS discharges as extensively documented in the literature [19,30,33,34]. These observations support the use of MD to probe the way by which HiPIMS-relevant parameters control transient interfacial structure, film morphology (including surface roughness), and residual defects that can degrade SRF performance.

While existing experimental studies provide crucial macroscopic correlations between deposition parameters and cavity performance, directly observing the transient, atomic-scale mechanisms during energetic plasma bombardment remains fundamentally challenging. Phenomena such as sub-nanosecond collision cascades, ballistic intermixing, and the rapid amorphization of the Nb–Cu interface occur on length and timescales that are largely inaccessible to in-situ experimental characterization, yet these transient events are the direct atomistic drivers of macroscopic film properties, including interfacial adhesion and void formation. This study, therefore, employs large-scale MD simulations as a dedicated, high-resolution computational complement to the existing experimental literature, isolating and probing these transient atomistic events to provide a mechanistic basis for the empirical optimization of SRF coatings.

In the present work, we use large-scale MD simulations to model the effect of the deposition conditions on the surface roughness, crystallization dynamics, microstructure, and interfacial structure of Nb films deposited on Cu substrates. By the deposition conditions, we understand the electrostatic bias on the substrate in terms of the energy of the atoms that arrive at the surface in the ionized state ($E_i$ ≈50–200 eV) and the fraction of ions in the deposition flux, which we refer to as the ionization fraction ($f_i$ ≈10%, 33%, 66%). Given the complex 3D geometry of elliptical SRF cavities, where different substrate regions are oriented at varying angles relative to the sputtering cathode, ranging from surfaces that directly face the cathode to those oriented nearly perpendicular to it, we employ a range of incident angles (0–75°) to capture the non-uniform flux distribution inherent to magnetron sputtering on such substrates. Our simulations show that an increase in ion energy and/or ionization fraction can drive transient surface roughening via rapid crystallization of a disordered interface layer. Moreover, we are able to correlate the interfacial composition with the amorphous structures reported for Nb-Cu alloys with very large miscibility gaps [35–37]. This observation allows us to assess the retained free volume, which eventually leads to the formation of single vacancies and vacancy clusters that potentially affect the film adhesion. The results reported herein provide atomistic insight that can contribute to and help rationalize the optimal bias voltage window observed experimentally. Rather than treating higher ion energy as universally beneficial, we conclude that process optimization should aim for controlled tuning of deposition parameters to target an interfacial composition near the identified optimum, a concrete, testable route for improving Nb-on-Cu SRF cavity performance.

## 2. Methods

Fig. 1(a) illustrates the macroscopic geometry of the HiPIMS setup within an SRF cavity. The coaxial configuration features a niobium cathode, support rod, and the copper cavity wall acting as the substrate. A localized region on the cavity, where the plasma sheath interacts with the wall, is selected for atomic-scale modeling.

MD simulations are employed to model Nb growth on a Cu substrate using the HiPIMS technique. The dimensions of the simulation box are approximately 10.3 × 10.3 × 5.5 $nm^3$, and the box contains around 34 000 Cu atoms arranged in a face-centered cubic (FCC) structure with the <1 1 1> direction parallel to the $z$ axis. Before depositing Nb atoms onto the Cu surface to model the deposition process, the simulation box is initially equilibrated at a constant temperature of 450 K using the NPT ensemble at 0 bar (Berendsen thermostat and barostat [38] for 60 ps). Periodic boundary conditions (PBCs) are applied in all three $x, y$, and $z$ directions. After this initial relaxation, the PBCs along the z-axis are removed, and the box is further relaxed for 10 ps. For this second step of relaxation, two atomic bottom layers (i.e., ≈ 3Å) are fixed to prevent cell movement during atomic deposition, and a thermal bath (NVT ensemble) within 5 Å is applied above the fixed bottom layers. This thermal bath is also applied on the side walls of the cell on the $x-z$ and $y-z$ planes to account for the thermal conductivity of the metal film-substrate system. The rest of the box follows the NVE ensemble, Fig. 1(b).

The HiPIMS technique generates a high-electron-density plasma that can effectively ionize the sputtered species up to 100%, with ion energies spanning within tens of eVs. At the same time, the degree of ionization of the sputtered atoms can be tuned by changing the process conditions, including the energy and width of the pulse [39,40]. This allows, e.g., the microstructure of the film and the deposition rate to be controlled. In the simulations, our aim is to study the effect of

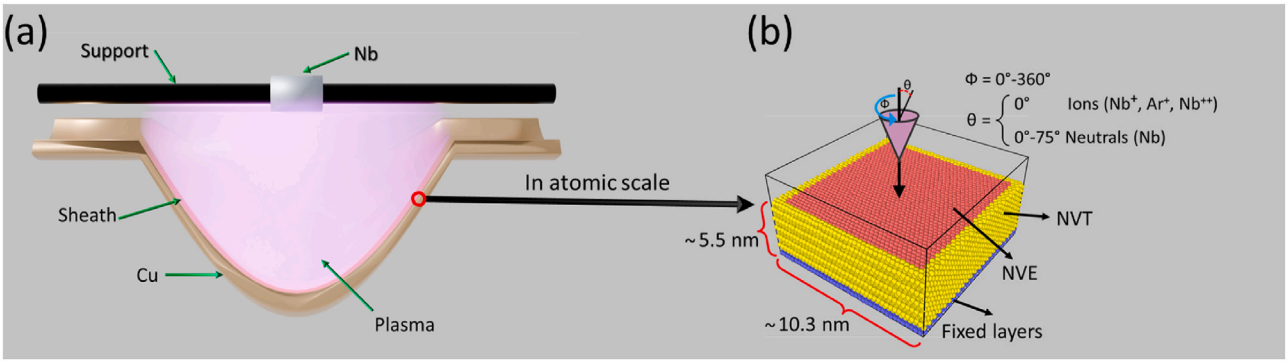



**Fig. 1.** Multi-scale modeling of Nb thin film growth for SRF applications. **(a)** Quarter cut of an SRF cavity, showing the support rod, the niobium (cathode), the copper cavity wall, and the plasma sheath region where deposition occurs. The red circle indicates the localized surface area modeled by MD. **(b)** atomistic simulation setup corresponding to the area marked in (a). The box dimensions are approximately $10.3 \times 10.3 \times 5.5$ nm$^3$, with two fixed bottom layers and NVT thermal bath zones to stabilize the system. The remaining atoms evolve under an NVE ensemble. Ionized particles ($Nb^+$, $Ar^+$, $Nb^{++}$) are introduced normal to the surface ($\theta = 0°$), while neutral Nb atoms are deposited at random angles between 0° and 75° to mimic realistic deposition conditions in RF cavity geometries. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

**Table 1**
Composition of the simulated ionization fractions. The table details the ratio (in %) of each species (Neutral Nb, $Nb^+$, $Nb^{++}$, $Ar^+$) for the three scenarios investigated, characterized by the fraction of ionized particles.

| $f_i$ | Neutral Nb% | $Nb^+$% | $Nb^{++}$ % | $Ar^+$% |
|---|---|---|---|---|
| 10% | 90 | 2.5 | 2.5 | 5 |
| 33% | 66 | 8.33 | 8.33 | 16.65 |
| 66% | 33.3 | 16.65 | 16.65 | 33.3 |

two key quantities: (i) the ionization fraction, $f_i$, and (ii) the energies of the ionized species $E_i$ that are controlled by the substrate bias voltage. The ionized species include $Nb^+$, $Nb^{++}$ and $Ar^+$ in equal proportions of the Nb and Ar ions, while the Nb ions are further equally divided into singly and doubly-charged ions, as shown in Table 1, and the total deposition flux consists of all species that arrive on the substrate during the deposition process, including neutral Nb atoms. Although $Nb^{++}$ requires higher excitation energies than $Nb^+$ and is generally less abundant in conventional plasmas, the exceptionally high peak electron densities characteristic of HiPIMS discharges produce significant fractions of multiply charged metal ions. We note that the inclusion in our simulations of doubly-charged ions is consistent with experimental observations in HiPIMS plasmas [41–43]. The equal $Nb^+/Nb^{++}$ ratio adopted here is therefore a controlled simplifying assumption that captures the aggregate physical effect of high-energy metallic projectiles without unnecessarily expanding the parameter space. These ions are included by increasing their impact energy twice compared to the singly charged ions of the same species.

With respect to $f_i$, we consider three scenarios: 10%, 33%, and 66% (low, moderate, and high ionization conditions). These specific fractions were chosen to represent the typical range of ion-to-neutral flux ratios practically achievable in HiPIMS discharges under varying process conditions, such as pulsing frequency, pulse width, and peak target current [39,40].

The ionized species are represented by Nb and Ar particles that are initialized with energies corresponding to those of ions $Nb^+$, $Nb^{++}$, and $Ar^+$ and arrive at the substrate with the energy according to the bias applied to the substrate. The remainder of the deposition flux (90%, 66%, and 33%) corresponds to sputtered neutral Nb atoms that approach the substrate surface with thermal velocities, modeled by assigning 1 eV energy to the neutral Nb atoms. Because MD simulations operate on the picosecond-to-nanosecond timescale, the temporal profile of the deposition flux, which is on the order of 100s of $\mu$s [44,45], cannot be explicitly described. Instead, the systematic variation of $f_i$ (10%, 33%, 66%) serves as a physically motivated representation of the dynamically evolving ratio of metal ions to Ar ions and neutrals that characterizes different temporal windows of the active discharge phase.

Energetic projectiles are injected with near-normal incidence to approximate sheath focus, while neutral Nb atoms are inserted with incident angles randomly sampled from 0° to 75° relative to the normal surface to mimic an elliptical (U-shaped) cavity geometry, see Fig. 1 [46].

The energies of the ionized species, i.e., energetic projectiles, $E_i$, are also varied to model the effect of substrate bias: singly charged species are assigned $E_i$ values of 50, 70, 80, 100, and 200 eV, while doubly charged $Nb^{++}$ projectiles are given the double the corresponding energy. However, for consistency, hereafter we refer to the substrate bias condition as the basic ion energy for a singly charged species, while implicitly assuming that the corresponding fraction of doubly-charged ions is also present in the depositing ion flux. Because the plasma potential contribution is typically only a few volts, the incident kinetic energy $E_i$ in our simulations corresponds directly to the applied substrate bias voltage used in experimental HiPIMS setups, and the two quantities are very nearly equivalent under the conditions studied.

For deposition, each incoming particle is added to the simulation box with a 0.6 ps interval and a distance of 1 nm above the surface. This distance is selected to be twice the largest cutoff among the interatomic potentials used. The total number of deposited particles corresponds to a coverage $\Theta$ of 22 monolayers (ML), equivalent to ≈4.6 nm, and the deposition simulation is performed at a temperature of 450 K. Substrate temperature was held constant at 450 K to isolate the specific effects of plasma bombardment energy and ionization fraction, as the influence of substrate temperature on HiPIMS-grown Nb film microstructure, alongside deposition angle effects for DCMS, was systematically investigated in our previous work [19]. Each as-deposited simulation cell undergoes annealing at 750 K for 5 ns (in the NVT ensemble) with the purpose of investigating the effect of thermal treatment on film morphology and microstructure. Annealing is a common strategy used to reduce defect density, promote grain growth, and relieve stresses in thin films [47–49]. Subsequently, simulation cells with the annealed structures are cooled to room temperature (300 K) at a rate of $1 \times 10^{11}$ K/s and then thermalized for an additional 15 ps. To ensure statistically robust characterization, all final analyses are performed by averaging the results over 11 snapshots collected every 1.5 ps during this final thermalization period. Furthermore, to account for stochastic variations inherent in the deposition process, the simulation is repeated for three independent

samples for each set of deposition conditions. The final values presented in this work are given by the mean and standard deviation of these three independent simulations.

Simulations are performed using the LAMMPS software package [50]. Cu-Cu, Nb-Nb, and Cu-Nb interactions are described by the interatomic potentials developed by Mishin [51], Ackland [52], and Zhang [53], respectively, using the embedded-atom method (EAM) formalism. This specific cross-interaction potential overcomes the limitations of earlier models by reproducing a fully miscible liquid phase, which is essential for realistically simulating forced atomic mixing within thermal spikes at the interface. While quantitative absolute predictions near the eutectic composition require appropriate caution, this potential excels in describing non-crystalline states and natively predicts nanoscale compositional patterning at a length scale of ~2.3 nm, successfully capturing the clustering characteristic of amorphous Cu–Nb systems [53]. The chosen potentials have been validated to reproduce the static bulk properties and surface reconstructions of the Nb-Cu system [19,54,55], ensuring a reliable simulation of the film growth dynamics. Short-range repulsive interactions between the metal atoms are modeled by the Ziegler, Biersack, and Littmark (ZBL) universal potential, combined with the EAM potential. Ar-Cu/Nb interactions use the DMol potential [56]. The repulsive ZBL potential models short-distance repulsive interactions during collision cascades, initiated by the accelerated ionized particles in the substrate. The electronic excitation effects during impact are included as an additional friction force acting on energetic atoms.

Visualization and post-simulation analysis are performed using OVITO software [57]. Post-simulation analysis focuses on characterizing the film structure and interface using Partial Radial Distribution Functions (PRDFs) constructed for all atom-type pairs in the simulation box and the identification of amorphous regions using the Common Neighbor Analysis (CNA) method. CNA classifies each atom's local environment by analyzing the bonding topology with its nearest neighbors. To study the film surface topography, we construct a surface mesh modifier and determine the root mean square roughness, $\sigma$, calculated according to the following formula:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (Z_i - \overline{Z})^2}{N}}, \quad (1)$$

where $Z_i$ is the $z$-coordinate (height) of the $i^{th}$ atom, $\overline{Z}$ is the mean height of all atoms, and N is the number of surface atoms.

To analyze the formation of an amorphous layer at the film-substrate interface, we perform additional MD simulations on Cu-Nb alloys of various compositions: $Cu_xNb_{1-x}$, where $x$ ranges from 0.40 to 0.65. Amorphous samples are prepared by quenching from 2800 K (700 K higher than the melting point of $Cu_{50}Nb_{50}$) to room temperature, with a cooling rate of $5 \times 10^{10}$ K/s. The average atomic volume of these amorphous alloys is then compared with that of their fully crystallized and optimized counterparts. The crystalline state is achieved by first minimizing the structure, thermalizing it for 1 ns at 1000 K using an NPT ensemble, and then applying a slow cooling rate of $5 \times 10^{10}$ K/s. To reach thermodynamic equilibrium, we apply the Monte Carlo molecular dynamics optimization algorithm (MCMD) [58] every 50 K for 10 ps and 1000 swaps.

## 3. Results

### 3.1. Surface roughness and crystallization dynamics

In this study, we follow the evolution of surface roughness and degree of crystallinity of the deposited films in the simulation runs under different deposition conditions. These conditions include the three different ionization fractions $f_i$ (the ionized flux consists of both high-energy Nb and Ar species) and the energies of Nb and Ar atoms, $E_i$, arriving at the substrate surface and controlled by the voltage bias applied to the substrate. The subsequent analysis of the thin film properties and their correlations with the deposition conditions after additional annealing provides further insights into the thermal stability of the film-substrate system in question. First, we detail these findings, with a particular focus on crystallization during the initial growth stages, because it directly affects surface roughness and, consequently, the performance of SRF cavities.

In the upper panels of Fig. 2(a, c, and e), we show the evolution of the surface roughness parameter, $\sigma$, calculated according to Eq. (1), serving as a continuous quantitative measure of roughness evolution as a function of deposited film thickness. The bottom panels of Fig. 2(b, d, and f) show the corresponding fraction of atoms in a BCC environment with 8 nearest neighbors of the total number of Nb atoms deposited, $f_{BCC}$: $f_{BCC} = \frac{N_{Nb(BCC)}}{N_{Nb}}$. Both parameters are shown as functions of the coverage $\Theta$ for different simulation conditions. In Fig. 2(a–e), we observe a consistent pattern: In all the cases studied, $\sigma$ shows a distinct peak at $\Theta \approx 3$ ML, followed by a slight decrease before transitioning to a monotonic, energy-dependent increase.

When the deposition flux is only slightly ionized, ($f_i = 10\%$), $\sigma$ increases with $\Theta$ nearly linearly, see Fig. 2(a). The lower the energy of the ions in the flux, the higher the value of $\sigma$ is at the end of the simulation ($\Theta \approx 22$ ML). The only exception is observed for the energy of the deposited ions $E_i = 200$ eV; in this case, $\sigma$ grows nonlinearly with $\Theta$, saturating at the value of $\approx 3$ Å, which is the lowest roughness obtained for the flux with $f_i = 10\%$.

The $f_{BCC}(\Theta)$ functions exhibit sigmoidal shapes, see Fig. 2(b), with nearly zero values at the beginning of the deposition followed by sharp increases and saturation at the end of the simulations. In addition, the saturation values of $f_{BCC}$ decrease from 79.5 to 73.7% when $E_i$ increases from 50 to 200 eV. More notably, the onset of the sharp increase for $E_i = 200$ eV occurs at $\Theta \approx 4$ ML vs. $\Theta \approx 0.5$ ML for all other conditions.

At low ion fractions ($f_i = 10\%$, Fig. 2(a)), the trend where a higher ion energy leads to smoother films aligns with the general understanding in the literature [39,59] that energetic bombardment in the range of several 10s of eV to ~100 eV promotes the mobility of adatoms, favors layer-by-layer growth and facilitates filling of troughs without causing significant surface damage [60–63]. Under these conditions, the mobility of adatoms (surface atoms that have not yet become a part of the continuous surface layer) dominates the morphological evolution throughout the deposition, resulting in monotonic behavior of the roughness parameters. However, even at this low ionization fraction, the observation that excessive energy ($E_i = 200$ eV) suppresses crystallization in the film (Fig. 2(b)) provides an early indication of competing mechanisms that become more pronounced at higher ion fractions.

Fig. 2(c) presents the evolution $\sigma(\Theta)$ observed in the simulations with moderately ionized deposition flux, i.e., $f_i = 33\%$. For the energy range $E_i = 50$–$100$ eV, $\sigma$ increases monotonically until $\Theta \approx 13$ ML, after that the increase in $\sigma$ becomes less pronounced, saturating at $\sigma \approx 1.9$Å when $\Theta \approx 22$ ML. In contrast, at $E_i = 200$ eV, the surface remains fairly smooth until $\Theta \approx 9$ ML, after which $\sigma$ increases rapidly within the $\Theta$ range 9 to 14 ML, reaching values above the roughness obtained for all other simulation conditions and eventually saturating at $\sigma \approx 2.2$Å. This behavior suggests that, at this ionization fraction, a high deposition energy induces either a smoothing or roughening effect on the growing surface in a seemingly random manner.

The evolution of $f_{BCC}(\Theta)$ for $f_i = 33\%$, presented in Fig. 2(d), is qualitatively similar to that seen for $f_i = 10\%$ (Fig. 2(b)). However, careful inspection of the curves shows that the onset of crystallization (as manifested by a sharp increase in $f_{BCC}$) continuously shifts from $\Theta \approx 1$ ML at $E_i = 50$ eV to $\Theta \approx 8$ ML at $E_i = 200$ eV. For the latter case, the onset of crystallization coincides with the increase in roughness observed in Fig. 2(c).

In the Supplementary material, the two videos show the time-resolved evolution of surface roughening, which is demonstrated by surface mesh visualization during film deposition simulations at $f_i = 33\%$ and $E_i = 50$ eV (Movie 1) and $E_i = 200$ eV (Movie 2). In both cases, the

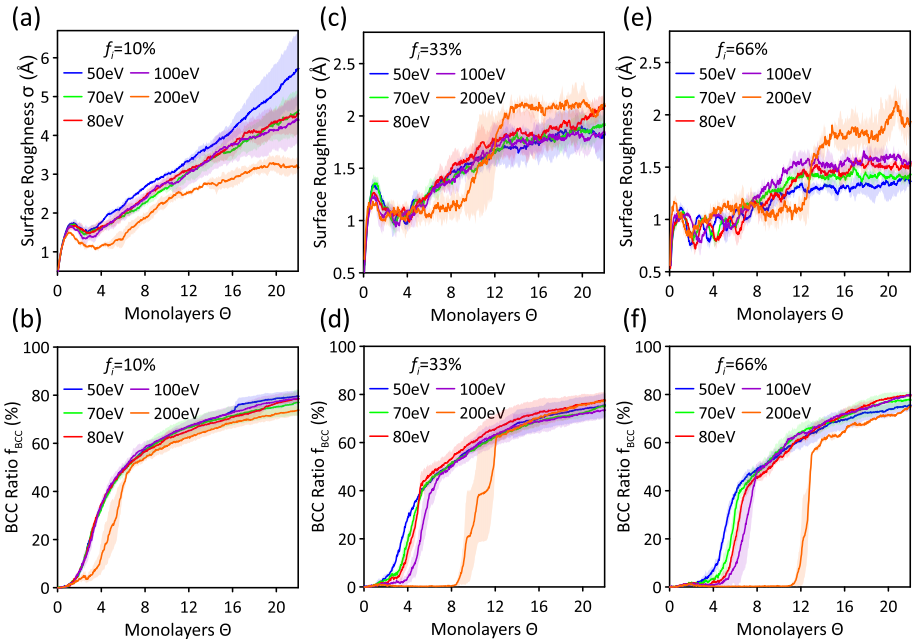



**Fig. 2.** Evolution of surface roughness parameter $\sigma$, Eq. (1)(a, c and e) and the BCC atomic fraction $f_{BCC} = \frac{N_{Nb(BCC)}}{N_{Nb}}$ (b, d and f) during the deposition simulations of Θ = 22 ML of Nb on a Cu substrate under different deposition conditions. The panels are organized according to different ionization fractions $f_i$, while the ion energies $E_i$ are color-coded in the legends. The shaded regions represent the standard deviation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

color bars show the same range of values between the lowest and highest coordinates of the surface mesh found in both simulations. In the right-hand bottom corner of the videos, the plots show the evolution of the $f_{BCC}$ fraction as a function of the surface coverage Θ. The running red circle follows the time elapsed which corresponds to the time evolution in the corresponding simulation cell. These evolutions reveal the temporal correlation between the formation of the BCC structure and the surface morphology evolution. At lower ion energy ($E_i$ = 50 eV), the surface features grow monotonically throughout the deposition process. In contrast, at high ion energy ($E_i$ = 200 eV), the surface remains flat until Θ ≈8 ML. After the onset of crystallization, the roughness increases rapidly during the deposition of the next ≈5 ML (Θ = 8–13 ML), ultimately producing localized height variations of 11–12 Å by Θ ≈22 ML. This rapid increase in roughness in the range Θ = 8–13 ML coincides with the sharp increase in $f_{BCC}$.

BCC formation animations reveal fundamentally different crystallization pathways. At $E_i$ = 50 eV (Movie 3), multiple BCC nuclei form early (Θ ≈2 to 3 ML) and grow gradually throughout the deposition process, while at $E_i$ = 200 eV (Movie 4), crystallization is suppressed until Θ ≈8 ML, then proceeds rapidly with anisotropic propagation: vertical growth (in the growth direction of the film) of localized BCC nuclei is significantly faster than in the lateral directions.

At $E_i$ = 200 eV ($f_i$ = 33%), a qualitatively different behavior emerges compared to what is observed at lower energies or at $f_i$ = 10%. At $E_i$ = 50–100 eV, the nucleation of BCC begins immediately (Θ ≈1 ML), and the roughness accumulates gradually. At $E_i$ = 200 eV, a two-stage behavior is observed. In stage 1 (Θ <8 ML), the surface remains smooth ($\sigma \approx$ 1Å) while $f_{BCC}$ remains close to zero. In stage 2 (Θ ≥8 ML), rapid crystallization is simultaneously accompanied by similarly rapid surface roughening. Moreover, in the video, which shows the temporal correlation between the onset of BCC crystallization and surface roughening (Movie 4 in the Supplementary Material), we see that crystallization under $f_i$ = 33%, $E_i$ = 200 eV deposition conditions drives the morphological evolution as well. At this ionization fraction, increasing $E_i$ from 50 to 200 eV progressively delays crystallization from Θ ≈1 ML to Θ ≈8 ML. This mechanism differs fundamentally from the monotonic roughening observed at $f_i$ = 10%, where adatom mobility controls morphology throughout deposition.

The amorphous Nb surface is characterized by a lack of long-range order and well-defined crystallographic features. Hence, Nb adatoms encounter a uniform potential energy and replicate the underlying surface structure as the film growth propagates. Upon crystallization, the Nb surface features well-defined (110) terraces bounded by steps. Our simulations show that steps effectively trap adatoms on terraces, which increases the top-layer nucleation probability and leads to the formation of three-dimensional surface morphology and an increase in surface roughness upon film thickening, similar to what is observed in kinetically-limited homoepitaxial metal-on-metal growth [64]. This is consistent with reported barriers for Nb on Nb(110) terrace diffusion and step-edge crossing [65].

Simulation data for the high ionization fraction ($f_i$ = 66%), see Fig. 2(e), show that the evolution of σ(Θ) is qualitatively similar to that obtained for $f_i$ = 33% (Fig. 2(c)): values of $E_i$ in the range 50–100 eV produce monotonic increases of $\sigma$ until it saturates at the plateau value of ≈ 1.5Å. However, deposition at $E_i$ = 200 eV again results in a sharp increase within the Θ range of 12–14 ML and a subsequent saturation to a value of $\sigma \approx$ 1.9Å. Comparison of data across different values of $f_i$, see Fig. 2 (a, c, and e) shows that an increase in the ionization fraction of the deposition flux leads to overall smoother surfaces. The evolution of $f_{BCC}$(Θ) for $f_i$ = 66% is plotted in Fig. 2(f). Again, a shift in the onset of crystallization with the energy is observed from Θ = 4 ML ($E_i$ = 50 eV) to Θ = 12 ML ($E_i$ = 200 eV), while the onset of crystallization and the sharp increase of $f_{BCC}$ at $E_i$ = 200 eV coincide with the increase in surface roughness in Fig. 2(e).

At high ion fractions ($f_i$ = 66%, Fig. 2(e and f)), the same two-stage mechanism persists, but with further delay in crystallization (from Θ = 8 ML at $f_i$ = 33% to Θ = 12 ML at $f_i$ = 66% for $E_i$ = 200 eV). This systematic delay in crystallization onset with increasing $E_i$ and $f_i$ results from competing mechanisms that are quantified through structural analyses in subsequent sections.

At $E_i$ = 200 eV, the ionized species impacting on the surface trigger small collision cascades, densifying the film but also forcing some atoms

into non-equilibrium positions. Meanwhile, they also affect the film composition through ion mixing. Direct quantification of the atomic content in 10.3×10.3×1 $nm^3$ volumes within the film (away from the interface) immediately before the appearance of crystallization reveals systematic compositional differences. At $f_i = 33\%$, $\Theta \approx 7$ ML, the region contains 5032±28 Nb, 1249±23 Cu, and 71±2 Ar atoms (averaged over three independent simulations), compared to 5271±47 Nb, 891±65 Cu, and 107±6 Ar atoms at $f_i = 66\%$, $\Theta \approx 11$ ML. Overall, the higher $f_i$ produces the films with 5% more Nb content, 29% less Cu intermixing, and 51% more Ar incorporation.

Fig. S1 plots the change of $\sigma$ ($\Delta\sigma$) and $f_{BCC}$ ($\Delta f_{BCC}$) upon annealing. The results show that the annealing process leads to smoother films ($\Delta\sigma < 0$); more negative values $\Delta\sigma$ correspond to a stronger smoothing of the surface morphology. In addition, annealing promotes crystallization ($\Delta f_{BCC} > 0$) under all the conditions used to simulate film growth. For the remaining analyses, annealing of as-deposited simulated films is performed at 750 K. It should be noted that by annealing we understand ~ 5 ns MD simulation runs at elevated temperature, which lead to surface smoothing caused by short-range atomic surface relaxation of metastable defects, e.g., strongly undercoordinated surface atoms, that appeared due to energetic deposition. This process is distinct from macroscopic grain growth and surface faceting, which operate on experimental timescales orders of magnitude longer and typically produce surface roughening.

### 3.2. Partial radial distribution function analysis

To understand the structural order and interfacial composition of the grown film, we further analyze the partial radial distribution functions (PRDFs) for specific atomic pairs. This approach characterizes the structure of the grown film (Nb-Nb), the degree of interfacial mixing (Nb-Cu), and the incorporation of processing Ar gas (Ar-Cu). Fig. 3 shows the PRDFs for all pairs in the simulation box, i.e., Nb–Nb (a-c), Nb–Cu (d-f), and Ar–Cu (g-i) for all deposition ion energies, $E_i$, and ionization fractions $f_i$. In these plots, we focus only on the first peak to quantify the short-range order of the film.

The Nb-Nb PRDFs for the lowest $f_i$ parameter look very similar for all ion energies $E_i$ as can be seen in Fig. 3(a). This result indicates that the ion energy does not influence the short-range structural order of the film at this low ionization fraction. The situation changes when the ionization fraction increases. From Fig. 3(b and c), the first peak of Nb-Nb PRDF stands out strongly when the ion energy is the highest ($E_i$ = 200 eV). Moreover, for the highest ionization fraction, $f_i = 66\%$, we see that the peak height gradually increases with increasing $E_i$ from 50 to 200 eV, indicating the strongest effect of the ion energy with this value of the ionization fraction.

The increase in the Nb-Nb PRDF peak height at $f_i = 66\%$ is consistent with atomic peening during film growth assisted by energetic ion beams, since energetic ions transfer the momentum in the direction

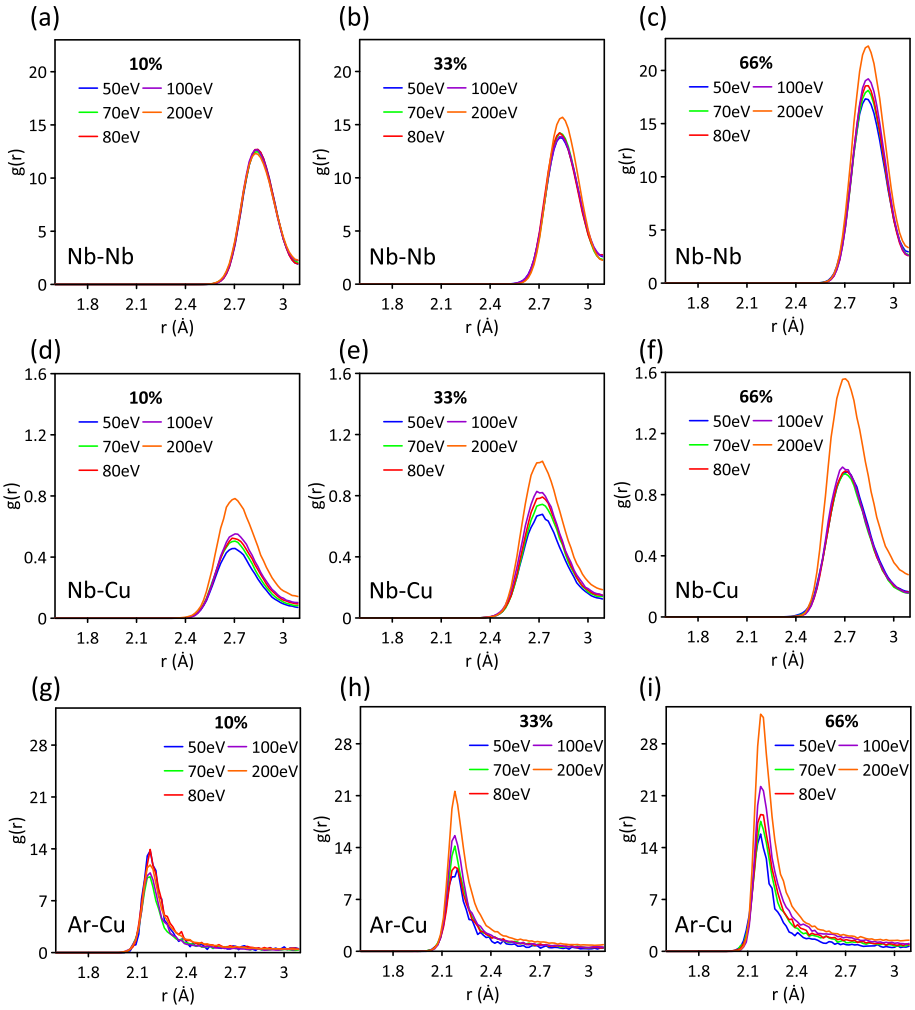



**Fig. 3.** Partial Radial Distribution Functions (PRDFs) characterizing the atomic-scale structure as a function of $f_i$ and $E_i$. **(a-c)** Nb-Nb correlations reveal that higher $f_i$ and $E_i$ lead to increased film ordering and densification. **(d-f)** Nb-Cu correlations show that intermixing at the interface is enhanced by both higher $f_i$ and $E_i$. **(g-i)** Ar-Cu correlations provide direct evidence of $Ar^+$ implantation, which increases significantly with higher $f_i$ and $E_i$.

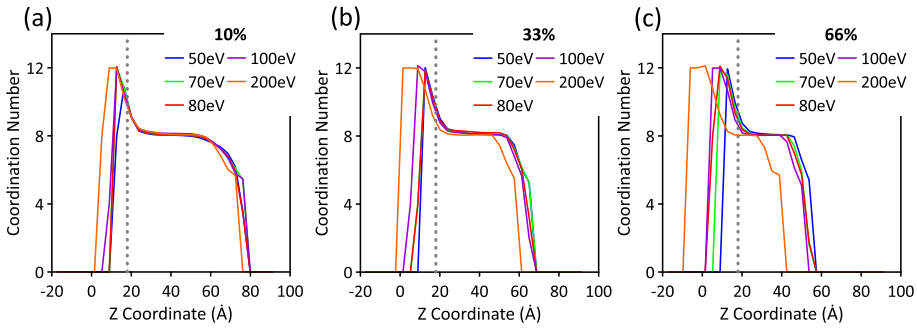



**Fig. 4.** Mean coordination number (CN) of niobium atoms as a function of depth (Z Coordinate) for the three simulated flux compositions: (a) 66%, (b) 33%, and (c) 10% ionized particles. Different colors represent different incident ion energies, as indicated in the legend. The gray dashed line shows the initial top surface position of the pristine Cu substrate. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

toward the substrate, forcing arriving atoms into more compact lattice positions with reduced excess volume, as reported in ion-bombarded systems with similar energy regimes [66–68]. This atomic-level densification mechanism explains the smooth conformal surfaces observed during Stage 1 growth at $f_i = 33$–$66\%$, $E_i = 200$ eV (Section 3.1): intense bombardment yields a compact film, but without lattice ordering, which is evident from the slow growth of the fraction $f_{BCC}$ during the early stage of deposition. Compare the videos in the Supplementary Materials, Movie 1 for $E_i = 50$ eV and Movie 2 for $E_i = 200$ eV at $f_i = 33\%$.

The Nb-Cu PRDFs, Fig. 3(d–f), quantify the atomic intermixing at the film-substrate interface. We see that at all $f_i$ values, the highest deposition energy resulted in a greater number of these bonds, indicating stronger atomic mixing. Moreover, the gradual growth of this peak from the lowest energy of the depositing ions to the highest is observed at both values of $f_i$ of 10% and 33%, while at $f_i = 66\%$, the height of the peak does not show dependence on the substrate bias (the energy of ions) until the energy is 200 eV. The sudden growth of the peak at this energy indicates that the dynamics of film growth at this substrate bias ($E_i$) and the ionization fraction ($f_i$) are different and lead to strong atomic mixing during this deposition condition.

This behavior can be explained by strong ballistic mixing and subplantation [69]. The monotonic increase in the Nb-Cu peak with $E_i$ and $f_i$ corresponds to the formation of a compositional gradient at the interface that extends deeper into the substrate. Higher ion energy and flux create more extensive interfacial mixing, disrupting epitaxial templating from the Cu substrate.

The Ar–Cu PRDFs, Fig. 3(g–i), quantify processing gas incorporation at the interface. Overall, we see that the number of Ar atoms trapped in the Cu layer is not very high; however, a similar tendency to that observed for other PRDFs indicates that during deposition at the substrate bias of 200 V (ion energy $E_i = 200$ eV), the Ar ions penetrate the deepest into Cu, forming more of the Ar-Cu pairs than at other ion energies. Moreover, the growing trend is seen for both values of $f_i$ (33% and 66%), while the lowest ionization fraction of 10% shows only insignificant penetration of the Ar atoms into the Cu substrate for all ion energies.

The extensive penetration of Ar ions into Cu leads to additional atomic dynamics during film growth. Since Ar atoms do not bind chemically in the matrix, the large amount of these atoms in the interface is likely to lead to the formation of gas bubbles as well as the nucleation of dislocations [33] storing elastic energy at the interface. This stored elastic energy drives the rapid crystallization and surface roughening observed at $\Theta = 8$ ML ($f_i = 33\%$, $E_i = 200$ eV) and $\Theta = 12$ ML ($f_i = 66\%$, $E_i = 200$ eV) (Section 3.1). The delayed formation of BCC nuclei was observed at the interface. The correlation between Ar incorporation (quantified by Ar-Cu PRDF peak heights of 32 at $f_i = 66\%$, $E_i = 200$ eV) and delayed crystallization onset ($\Theta = 12$ ML under these conditions) demonstrates that processing gas implantation contributes to the energy barrier that must be overcome to complete crystallization at the interface.

### 3.3. Effect of Nb coordination number and atomic mixing at the interface on quality of grown thin films

To further analyze the evolution of the BCC lattice crystallization onset and the atomic packing density at different ion energies, we plot in Fig. 4 the depth profile of the Nb coordination number (CN) for all values $E_i$ and $f_i$. In these graphs, the beginning of the Z-axis is shifted beneath the original surface by ≈20 Å to enable tracking of the penetration dynamics of the deposited atoms. The original surface is indicated by the vertical dashed gray lines.

At $f_i = 10\%$, (Fig. 4(a)) we see that CN > 0 for Z coordinate reaching values of ≈80 Å. This is larger compared to all other $f_i$ values (panels (b) and (c)), indicating that the thickness of the film under this condition reaches its maximum. Moreover, this thickness does not depend on the energy of the incoming ionized atoms in the deposited flux. However, we see that, at the highest ion energy, the Nb atoms penetrate deeper beneath the original surface (roughly 20 Å) than at any other energy values. In addition, these atoms incorporated into the Cu substrate tend to have a higher CN (CN = 12), similar to that of the Cu atoms. At $f_i = 33\%$ (Fig. 4(b)), the film thickness decreases slightly (maximum Z for which CN > 0 in the range 60 to 65 Å. Also, we observe that at the highest ion energy (200 eV), the film becomes compact, as evidenced by the decrease in the film thickness and the deeper penetration of Nb atoms into the Cu substrate. At $f_i = 66\%$ (Fig. 4(c)), the Nb atoms penetrate the deepest into the Cu substrate (up to 30 Å), resulting in the minimum film thickness. Moreover, at this ionization fraction, we see that the thickness decreases gradually with an increase in $E_i$. This systematic decrease in film thickness at the highest incident energies is consistent with the onset of a sputtering-dominated regime, sometimes referred to as an “etching” mode, in which intense ion bombardment actively removes deposited material from the surface at a rate that partially offsets the deposition flux.

The deep penetration of Nb atoms leads to the formation of the intermixed layer at the Cu-Nb interface, whose width increases with $E_i$. At $f_i = 33\%$, the depth of penetration of Nb and the width of the mixed layer (≈25 Å at $E_i = 200$ eV) are smaller compared to those obtained at $f_i = 66\%$ with those same $E_i$ values (≈27 Å at $E_i = 200$ eV). At $f_i = 10\%$ and $E_i = 50$ eV, Nb mixing is confined to the region of immediate interface with minimal penetration of Nb under the Cu surface, producing the narrowest width of the intermixed interface among all the conditions studied.

The systematic increase in the intermixed width around the interface results from ballistic sub-plantation and collision cascade mixing [70–72]. The reduced penetration at lower $f_i$ reflects the reduced energy flux of the ionized species, which reduces the intensity of the collision cascades. This depth-dependent CN analysis quantifies the ballistic

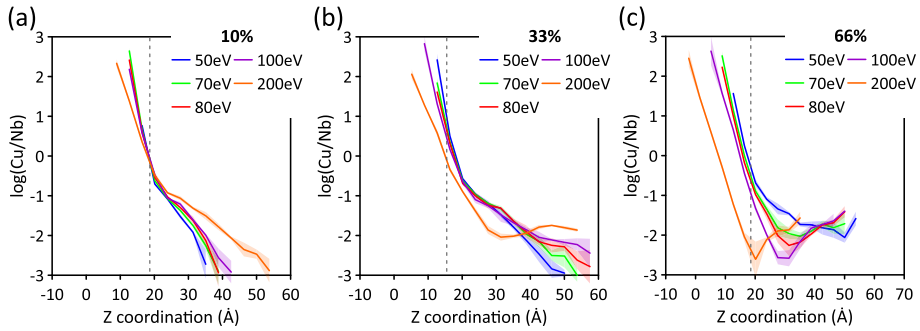



**Fig. 5.** Analysis of compositional gradient plotted as the logarithmic Cu/Nb concentration profiles at Θ ≈22 ML for $f_i$ equal to (a) 10%, (b) 33%, and (c) 66%, respectively. Within each panel, different colored lines denote simulation runs for different $E_i$ values for singly ionized species from 50 eV to 200 eV, as indicated in the legend.

mixing mechanism inferred from the peak heights of Nb-Cu PRDF in Section 3.2. The increasing intensities of the Nb-Cu PRDF peak with $E_i$ and $f_i$ directly correspond to a deeper penetration of Nb and wider intermixing regions around the interface.

The penetration of Nb into the Cu substrate implies that there may also be Cu atoms mixed into the Nb thin film, which we have reported in Ref. [19]. To quantify the compositional gradient across the interface, we analyzed the variation of the Nb-to-Cu content (plotted as log(Cu/Nb)) as a function of the Z-coordinate for all conditions in the final deposition state, see Fig. 5. In these graphs, the positive values of the logarithm function show the excess of Cu atoms over the Nb ones, while the negative values indicate the mixing of the Cu atoms into the film. In these graphs, the beginning of the Z-coordinates is chosen the same as in Fig. 3, which is again shown as the dashed gray lines. At $f_i = 10\%$ (Fig. 5(a)), log(Cu/Nb) exhibits two distinct decay rates: a steep slope in the Cu-rich region (i.e., little mixing), which changes to a less steep slope in the Nb-rich region, indicating a more efficient mixing of Cu atoms in the Nb film. Moreover, the lowest slope is seen for $E_i$ = 200 eV, which means that even at the lowest ionization fraction, the high energy of the incoming ions promotes the mixing of Cu into the Nb film more strongly than the penetration of Nb atoms into the Cu substrate.

At $f_i = 33\%$ (Fig. 5(b)), we observe a similar trend of the two-slope profile for energies below 200 eV. At $E_i = 200$ eV, the profile becomes non-monotonic: log(Cu/Nb) decreases through the interface with a slope similar to that of the Cu region, but at a depth of 30 Å it reaches the minimum and then slightly increases to a saturated value that remains until the surface. This trend indicates Cu contamination at the surface. At $f_i = 66\%$ (Fig. 5(c)), the two-slope trend practically disappears. Instead, the log(Cu/Nb) function begins much deeper beneath the surface (strong penetration of Nb into the Cu substrate) and decreases steadily until the minimum value within the film. After that, it increases rapidly toward the surface for all $E_i$. The depth of the minimum (where the film is the least contaminated with Cu) decreases with increasing $E_i$. At the highest energy, $E_i$ = 200 eV, this minimum lies almost at the interface, which means that under this condition, Cu mixes the most efficiently in the Nb film. At the lowest energy, $E_i$ = 50 eV, the slope of the log(Cu/Nb) function is somewhat shallower, which only indicates less dynamics; however, the incorporation of Cu atoms in Nb is still significant.

The transition from two-slope to single-slope log(Cu/Nb) profiles with increasing $f_i$ reflects different mixing mechanisms. At $f_i = 10\%$, the distinct slopes reflect compositional asymmetry across the interface: (i) the steeper gradient within the substrate shows limited Nb penetration at low ion energies and fluxes, creating a sharp compositional transition (the width of the interface changes from ≈ 7Å to ≈ 13Å when $E_i$ changes from 50 to 200 eV); (ii) the decrease of the gradient through the Nb film with increase in the $E_i$ shows that even at this low ionization fraction there is the Cu upward migration driven by collision cascade-induced atomic mobility and the thermodynamic forces as we showed in Ref. [19]. It is important to emphasize that this upward Cu migration is not driven by classical, long-range bulk thermal diffusion. Rather, it is activated by the highly transient atomic mobility within localized thermal spikes generated by individual collision cascades, allowing local thermodynamic driving forces to promote Cu segregation on the picosecond timescale. This mechanism distributes Cu throughout the film, producing an extended compositional tail. At $f_i = 66\%$, intense collision cascades homogenize the composition at regions, in the interface, and in the film, eliminating the two-slope transition by creating a single monotonic compositional gradient from within the substrate to the middle of the film.

The upward Cu migration is the dominant mechanism responsible for the surface composition during film growth under $f_i = 33\%$ and $E_i = 200$ eV, as well as for all ion energies under $f_i = 66\%$ conditions. At these deposition conditions, sputtering of surface atoms starts playing a role, which is evidenced by the decreasing film thickness with increasing $E_i$ (Fig. 4). In this process, both Cu and Nb are removed, and the preferential mobility of Cu atoms within collision cascades continuously replenishes the surface Cu concentration. The non-monotonic log(Cu/Nb) profiles (Fig. 5(b and c)), which show Cu depletion within the growing film, followed by surface enrichment, are the clear signatures of this upward Cu migration. Cu is extracted from within the film regions and transported to the surface, creating the characteristic depletion-to-enrichment profile that cannot be explained by sputtering alone.

### 3.4. Interface disorder under energetic HiPIMS deposition

We follow the structural evolution during the simulations of the growth of the Nb thin film under HiPIMS conditions using the common neighbor analysis (CNA) routine available in OVITO [57]. In our simulations, we consistently observe the formation of a disordered layer at the Nb/Cu interface. This is identified as a large fraction of atoms that are labeled by CNA as 'Other', because the local coordination of these atoms is not consistent with FCC-/HCP-Cu or BCC-Nb. The thickness of this layer at the interface is highly dependent on the deposition conditions: the higher the energy and the higher the ionization fraction, the thicker and more disordered the layer becomes, see Fig. 6. In this figure, we showcase the interfacial disorder for the lowest and highest values of $E_i$ and $f_i$. Here, the left panels (a,b) show the case of $E_i$ = 50 eV and $f_i$ = 10%, while the right panels (c,d) show the case of $E_i$ = 200 eV and $f_i$ = 66%. To emphasize the disorder and thickness of the disordered layer, we plot the atomistic models of the interfaces colored according to CNA (the panels a, c) and the corresponding Voronoi polyhedra [73] (the panels b, d). In the latter analysis, one can clearly see that the atoms identified by CNA as 'Other' indeed belong to a random structure in the interface region, since the Voronoi polyhedra in these regions are not aligned with either the top or the bottom crystal structures.

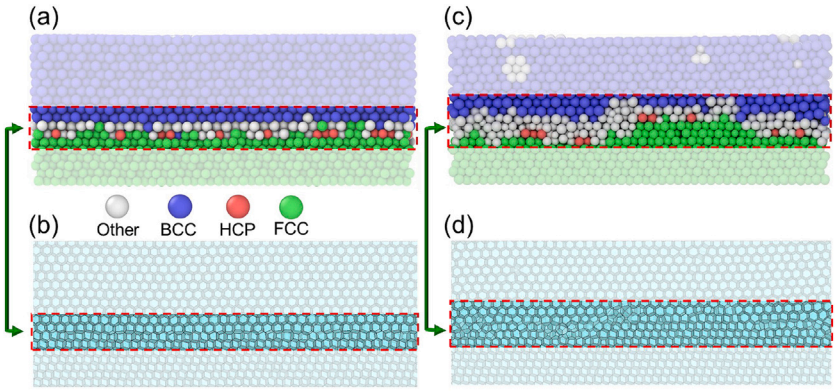



**Fig. 6.** Structure of the interface obtained during the thin film deposition with $E_i$ = 50 eV and $f_i$ = 10% in (a) and (c), and with $E_i$ = 200 eV and $f_i$ = 66% in (b) and (d). (a) and (b) show the atomic structure of the interface with the atoms colored according to CNA in OVITO. (c) and (d) show the Voronoi polyhedra distributions of the same interfaces shown in (a) and (b). The Voronoi polyhedra emphasize the level of disorder in the interface, although the film under the conditions shown in (b) and (d) is overall much denser.

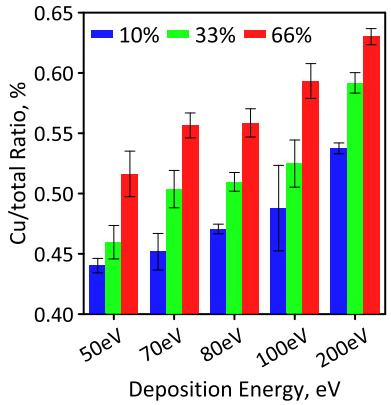



**Fig. 7.** Atomic Cu fraction in the disordered interface region as a function of deposition energy $E_i$ and ionization fraction $f_i$ of the deposition flux calculated as $N_{\mathrm{Cu('Other')}}/(N_{\mathrm{all('Other')}})$.

We also see that the stronger disorder is associated with the higher content of Cu atoms among the atoms marked by CNA as 'Other'. We relate the disordered structure to the amorphous structures of Cu-Nb alloys [35,36], and further analyzed the excess atomic volume in the disordered interface as compared to their crystalline counterparts. In Fig. 7, we plot the atomic percentages of Cu atoms with respect to the atoms in the interface marked as 'Other'. The results show that the amount of Cu in the disordered interface increases with increasing ion energy and ionization fraction. It is clear that the higher the number of Cu atoms that are mixed with Nb atoms, the stronger the disorder in the interface, which increases the elastic strain energy, promoting the formation of defects in the interface. Together, the high fraction of CNA "Other" atoms, the depth-resolved coordination number profiles (Fig. 4), and the Voronoi polyhedra distributions (Fig. 6, panels b and d) collectively indicate a region of strong disorder resembling an amorphous structure, as has been identified for similar compositions in Refs. [35,36]. In crystalline regions, Voronoi polyhedra adopt uniform, repeating geometric forms consistent with the underlying lattice symmetry; by contrast, the interfacial polyhedra are highly irregular and geometrically diverse. The disordered phase in our simulations appears in a very thin interfacial layer, which is why it is difficult to verify the short- and long-range orders; however, this geometric irregularity clearly confirms the lack of crystallographic alignment at the interface.

On the other hand, the Nb atoms that are ballistically inserted into an FCC coordination environment of Cu also become kinetically trapped in it. This potentially reduces nucleation of the BCC structure, contributing to a persistent amorphous interfacial layer that retains vacancy defects after deposition and annealing.

To analyze the excess atomic volume trapped in the disordered interface, we now turn to the bulk models of highly concentrated Cu-Nb alloys with the composition $Cu_xNb_{100-x}$ alloys, where x = 40–60 is measured in atomic percent. These structures are produced by quenching from the melt from 2800 to 300 K, see Section 2 for details. In the Supplementary Material, Fig. S2, we showcase the $Cu_{50}Nb_{50}$ alloy before and after the optimization using the MCMD method. As one can see, the quenched amorphous structure is fully phase separated into pure Nb and pure Cu crystal structures with minimal intermixing between these two layers. We calculate the excess free volume per 100 atoms using the following formula:

$$\Delta V_{100}(Cu_xNb_{100-x}) = (V^{d}(Cu) - V^{c}(Cu)) \times x + (V^{d}(Nb) - V^{c}(Nb)) \times (100 - x) \quad (2)$$

In Table 2, the results are normalized $\Delta V_{100}$ to an atomic volume of a single Cu atom in the FCC lattice ($V$(Cu) ≈ 11.95Å$^3$), producing the equivalent number of Cu-vacancies, which are likely to appear after the elastic strain relaxes.

The comparison of Fig. 7 and Table 2 reveals that the formation of the vacancies is expected practically under all the deposition conditions studied in this work. Our deposition simulations show that increasing deposition ion energy and ionization fraction consistently drives the interface toward Cu-rich composition (from 44 at.% Cu at $f_i$ = 10%, $E_i$ = 50 eV to 63 at.% Cu at $f_i$ = 66%, $E_i$ = 200 eV). As we see from Table 2, this leads to a higher number of vacancies that can form at the interface. However, this dependence is not very strong, which is likely explained by the denser packing of Cu around the Nb atoms with the increase in Cu content.

**Table 2**
Excess free volume and equivalent Cu-vacancy count ($N(Cu_{Vac})$) after crystallization of a $Cu_xNb_{100-x}$ alloy.

| $x$% | $\Delta V_{100}$ (Å$^3$) | $N(Cu_{Vac})$ |
|---|---|---|
| 40 | 18.62 | 1.56 |
| 45 | 18.41 | 1.54 |
| 50 | 19.00 | 1.58 |
| 55 | 19.04 | 1.59 |
| 60 | 19.68 | 1.65 |

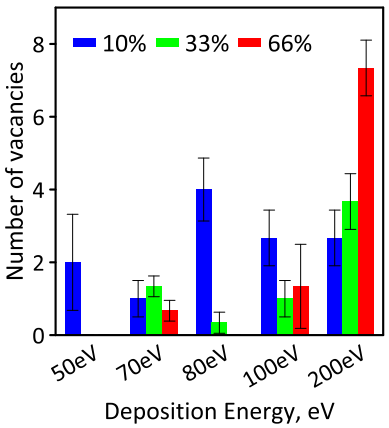



**Fig. 8.** Number of vacancies that appeared during the thin film growth in Nb away from the interface for all $E_i$ and $f_i$ studied in this work.

We also plot the number of vacancies in the Nb thin film away from the interface, see Fig. 8. Surprisingly, the number of these vacancies is not negligible, which may further affect the quality of the growing film at the high values of substrate bias and high ionization efficiency of the depositing flux.

## 4. Discussion

The surface morphology and roughness are critical parameters that affect the thin film performance, as shown in different studies of various films, including NbN [74] and Nb grown on the MgO(0 0 1) substrate [75], Cr coatings [76], and the thin film of Nb grown on amorphous $SiO_2$/Si(1 0 0) substrates [77]. For SRF cavities specifically, surface roughness directly influences the Q quality factor through multiple mechanisms: rough surfaces increase the enhancement of the RF magnetic field at protrusions, promote premature flux penetration, and create nucleation sites for thermal breakdown [78,79]. Moreover, interfacial defects at the Nb/Cu boundary can trap magnetic flux and degrade thermal conductivity, both of which contribute to the Q-slope phenomenon that limits cavity performance [9,80,81].

The initial transient peak observed in Fig. 2(a, c, and e) at $\Theta \approx 3$ ML is likely a computational artifact resulting from the low number of atoms used in the roughness calculation during the incipient stages of deposition, where the statistical analysis is insufficiently accurate. Beyond this peak, the evolution of the roughness reveals distinct growth regimes that depend strongly on both $f_i$ and $E_i$.

The atomic-level compositional analysis reveals that increased ion bombardment flux creates purer Nb films with reduced Cu intermixing but enhanced processing gas incorporation at the interface prior to crystallization. The structural characterization presented in Sections 3.2 and 3.3 shows the atomic-scale origin of this behavior. We propose that these competing mechanisms, i.e., atomic peening, ballistic mixing, and processing gas implantation, create a thicker disordered film with progressively higher stored elastic energy as $E_i$ and $f_i$ increase, ultimately governing the critical thickness at which rapid crystallization occurs.

The occurrence of this delayed crystallization at a critical thickness ($\Theta \approx 8$ ML for $f_i = 33\%$, $E_i = 200$ eV; $\Theta \approx 12$ ML for $f_i = 66\%$, $E_i = 200$ eV) is characteristic of structural transformations driven by size-dependent thermodynamic and kinetic factors. Studies of amorphous-to-crystalline [82–85] and BCC-to-FCC [86] transformations in thin films demonstrate that critical thicknesses arise from the balance between interfacial energy, strain energy, and bulk free energy differences. The rapid increase in roughness at $\Theta \approx 9$–14 ML coinciding with crystallization onset is consistent with such phase transformation-driven morphological instabilities.

The mechanisms governing crystallization can be seen in the analysis of the mixed-pair Ar-Cu and Nb-Cu PRDF (Section 3.2). At $f_i = 66\%$, $E_i = 200$ eV, the Ar-Cu peak height is twice as high as that observed for $E_i = 50$ eV, which indicates substantial $Ar^+$ implantation at the interface. This implanted Ar has previously been shown to create vacancy clusters and dislocations [33] that store elastic strain energy, contributing to the accumulation of interfacial stress. Furthermore, our observation that $Ar^+$ incorporation contributes to interfacial disorder and delayed crystallization provides a clear atomistic rationale for advanced deposition techniques such as selective ion acceleration [87], which are designed to minimize $Ar^+$ bombardment during film growth.

At the same time, the height of the Nb-Cu PRDF peak is the highest for the highest ionization fraction ($f_i = 66\%$) and the highest ion energy ($E_i = 200$ eV) compared to all other deposition conditions, indicating extensive ballistic mixing [69] that creates a compositionally inhomogeneous stressed interlayer that extends several nanometers into the substrate. Compositional gradient analysis using log(Cu/Nb) profiles at $\Theta \approx 22$ ML (Fig. 5) quantifies this interfacial composition further: the absence of the two-slope transition and the appearance of Cu surface enrichment for the deposition at $f_i = 66\%$ and $E_i = 200$ eV indicate that collision cascades have homogenized the interface while simultaneously driving upward Cu migration through the film. The transition from a two-slope profile (at low $f_i$) to a single-slope profile with surface Cu enrichment (at high $f_i$) suggests that intensive ion bombardment both widens the interfacial mixing zone and activates collision cascade-driven Cu segregation at the surface.

The crystallization of deposited films during the initial growth stages is significantly influenced by the ion energy and the ionization fraction of the deposition flux that arrives at the substrate, as well as the substrate temperature [88,89]. Adjusting these parameters can either accelerate or delay the crystallization process [90]. Our simulations show that the systematic increase in critical thickness from $\Theta = 1$ ML ($f_i = 10\%$, $E_i = 50$ eV) to $\Theta = 12$ ML ($f_i = 66\%$, $E_i = 200$ eV) reflects the accumulated energy barrier created by three synergistic mechanisms: (i) atomic peening-induced compressive stress, (ii) $Ar^+$ implantation-generated defect clusters, and (iii) compositional gradient that disrupts epitaxial templating. At $f_i = 33\%$, increasing $E_i$ from 50 to 200 eV progressively delays crystallization from $\Theta \approx 1$ ML to $\Theta \approx 8$ ML, demonstrating that energetic bombardment suppresses nucleation more effectively than thermal activation can promote it at the substrate temperature of 450 K. At higher $E_i$ and $f_i$, these mechanisms collectively create a thicker amorphous film with a greater stored elastic energy. The amorphous-to-BCC transformation is triggered when this stored energy reaches a critical thickness-dependent threshold where stress relief through crystallization becomes thermodynamically favorable, consistent with stress-relief-driven hillock formation observed in thin films under ion irradiation [91,92]. Ion-beam-induced defect migration [93] may facilitate this transformation once the critical thickness is reached, enabling rapid atomic rearrangement from the metastable amorphous structure to the equilibrium BCC phase.

The correlation between the appearance of crystallization and the increase in roughness under all high-energy conditions ($f_i = 33\%$, 66%; $E_i = 200$ eV) demonstrates that the phase-transition kinetics, rather than the mobility of adatoms, control the morphological evolution at high $E_i$ and $f_i$. We interpret the anisotropic transformation observed in surface-mesh animations, see Supplementary material, Movie 4 (vertical propagation significantly faster than lateral spreading), as resulting from preferential stress release in the growth direction, where the amorphous-to-BCC volume change is accommodated with minimal elastic constraint, compared to the lateral directions, which are constrained by the surrounding material. This anisotropic crystallization generates columnar BCC domains whose volumetric expansion drives out-of-plane displacement and substantial surface rearrangement [94], manifesting as the characteristic hillock morphology observed in Movie 4.

The decreasing final roughness with increasing $f_i$ (from 2.2 ($f_i = 33\%$) to 1.9 Å ($f_i = 66\%$) at $E_i = 200$ eV) despite similar

crystallization mechanisms and delayed transformation onset results from two interrelated factors. First, higher $f_i$ produces smoother Stage 1 amorphous surfaces through more effective atomic peening, providing flatter initial morphologies before transformation, as quantified by the lower pre-transition roughness values ($\sigma \approx 1.2$Å at $f_i = 33\%$ versus $\sigma \approx 1$Å at $f_i = 66\%$ before crystallization). Second, the thicker amorphous layer formed at higher $f_i$ (critical thickness increases from $\Theta \approx 8$ ML to $\Theta \approx 12$ ML) distributes the stress-relief displacement over a larger volume during transformation. While the total stored elastic energy is higher at $f_i = 66\%$, its release is spatially distributed rather than concentrated in a thinner layer, producing smaller localized surface perturbations and reduced hillock amplitudes. Thus, we conclude that while the crystallization mechanism remains fundamentally the same across $f_i = 33$–$66\%$ at $E_i = 200$ eV, the initial state from which transformation occurs, specifically, the pre-transition surface roughness and the thickness of the amorphous layer undergoing transformation, determines the final morphological outcome.

The morphological transitions and crystallization dynamics described in Section 3.1 are driven by atomic-scale structural changes at the film-substrate interface, as evidenced by the systematic correlations between interfacial composition (Fig. 5), coordination environment (Fig. 4), and crystallization onset ($f_{BCC}$ evolution in Fig. 2). These interfacial structural changes comprise two interrelated phenomena: (i) the kinetically-trapped amorphous film structure that delays BCC nucleation during Stage 1 growth, and (ii) the ballistically mixed amorphous interfacial layer whose composition and width are established during deposition and persist after film crystallization begins.

This mechanistic picture explains why the crystallization behavior changes qualitatively between $f_i = 10\%$ at $E_i = 50$–$100$ eV (monotonic roughening, adatom mobility-controlled) and $f_i = 33$–$66\%$ at $E_i = 200$ eV (two-stage mechanism with delayed crystallization and roughness spikes). Very energetic deposition at moderate-to-high ion fractions can delay ordering, then trigger rapid crystallization and morphological instability. At low $f_i$, the sharp compositional interface permits epitaxial templating from the Cu substrate, enabling immediate BCC nucleation. At high $f_i$, the homogenized compositional gradient disrupts epitaxial relationships, forcing the deposit into a metastable amorphous configuration.

Cu enrichment at the surface, combined with the FCC-coordinated Nb atoms at the interface and an excessive amount of implanted Ar that can result in defect clusters, see Section 3.2, collectively establishes the energy barrier that must be overcome at the critical thickness before crystallization can proceed. This Cu enrichment contributes to the delayed crystallization observed in Section 3.1 by inhibiting BCC nucleation through preventing the required Nb-Nb coordination and continuous depletion of the near-surface Nb concentration required for critical nucleus formation. The shift of the log(Cu/Nb) minimum to shallower depths with increasing $E_i$ at $f_i = 66\%$ quantifies the competing effects of deeper Nb implantation versus enhanced substrate erosion and Cu upward migration.

The compositional gradient profile created by the ion-induced mixing at higher values $f_i$ and $E_i$ enhances mechanical interlocking at the interface, which correlates with improved film adhesion [95–98]. However, we observe that this improved adhesion comes at the cost of creating a compositionally complex interlayer that delays crystallization, as demonstrated by the systematic shift of crystallization onset from $\Theta = 1$ ML ($f_i = 10\%$) to $\Theta = 12$ ML ($f_i = 66\%$) at $E_i = 200$ eV as seen in Figs. 2(d and f).

Under equilibrium conditions, Cu and Nb are immiscible over the entire composition range of the binary Cu-Nb system (positive heat of mixing, +4 kJ/mol [35]) and do not form alloys. Nevertheless, non-equilibrium processing (sputtering [99,100], and ion-beam mixing [35,36]) yields metastable amorphous Cu–Nb phases over a broad composition range, for example, between 15–72 at.% Nb [35] or 30–85 at.% Nb [36]. These amorphous phases exhibit surprisingly low formation enthalpies (4.5–7.6 kJ/mol [99]), which is explained by a sign change in the liquid-phase heat of mixing upon undercooling. Similarly, atom-probe tomography has further revealed that nanoscale phase separation, coexisting Cu-rich and Nb-rich clusters within the amorphous matrix, lowers the free energy below that of any supersaturated crystalline phase [37,101,102]. As-deposited amorphous Cu–Nb films likewise display high resistivity and composition-dependent superconducting transitions [100,101]. Upon annealing, these films devitrify in two stages: FCC-Cu nanocrystals form within an Nb-rich amorphous matrix, followed by the growth of BCC-Nb grains that encapsulate the Cu. Moreover, the short-range order of the parent FCC-Cu and BCC-Nb solids influences the resulting amorphous structure, a phenomenon termed “structural heredity” [36].

This compositional control through deposition parameters provides a direct mechanism for engineering the interfacial structure. The ballistic mixing and collision cascade processes that create the interfacial amorphous layer are strongly influenced by $E_i$ and $f_i$: increasing both parameters widens the interfacial mixing zone and drives the composition toward Cu enrichment (from 44 at.% Cu at $f_i = 10\%$, $E_i = 50$ eV to 63 at.% Cu at $f_i = 66\%$, $E_i = 200$ eV). Upon subsequent annealing or slow thermal relaxation, these interfacial layers crystallize and retain vacancy concentrations determined from separate equilibrium molecular dynamics simulations (MCMD) of bulk Cu-Nb amorphous alloys (Section 2) that depend non-monotonically on composition.

Interfaces formed at both extremes, i.e., Cu-poor conditions (low $E_i$, low $f_i$: 40 at.% Cu, approximately 1.56 equivalent Cu vacancies per 100 atoms) and Cu-rich conditions (high $E_i$, high $f_i$: 60 at.% Cu, approximately 1.65 equivalent Cu vacancies per 100 atoms), retain substantially higher vacancy concentrations compared to interfaces formed near the optimal composition ($Cu_{45}Nb_{55}$, approximately 1.54 vacancies per 100 atoms). This non-monotonic vacancy formation behavior explains a critical trade-off in the process optimization.

Our analysis suggests that at low $E_i$ and $f_i$, insufficient ballistic mixing creates narrow, Cu-depleted interfacial layers that generate excess vacancies due to poor atomic packing during crystallization. At high $E_i$ and $f_i$, while deep interfacial mixing produces well-adhered interfaces (Section 3.3), the excessive Cu enrichment creates compositionally suboptimal amorphous layers that similarly retain high vacancy concentrations after crystallization. Furthermore, as evidenced by the Ar-Cu PRDF analysis in Section 3.2, these high-Cu deposition conditions (high $E_i$, high $f_i$) are simultaneously accompanied by significantly enhanced $Ar^+$ implantation at the interface. This trapped Ar acts as a compounding factor, generating additional local strain and vacancy clusters beyond those predicted by the bulk free-volume analysis alone, further exacerbating the defect density in the Cu-rich regime. The minimum vacancy concentration occurs at intermediate conditions (approximately $f_i = 33\%$, $E_i = 70$–$100$ eV, Fig. 7), where the interfacial composition approaches $Cu_{45}Nb_{55}$ and atomic packing during crystallization most closely matches the equilibrium coordination environment.

These interfacial vacancies represent permanent structural defects that persist within the temperature range simulated (up to 750 K annealing), regardless of the film’s subsequent crystallization behavior. Minimizing these potential vacancy reservoirs is critical, as they are known to degrade SRF performance through multiple mechanisms: (i) trapping of residual impurities (e.g., H, Ar) introduced during operation, which locally suppresses superconductivity [103,104], (ii) acting as pinning sites for trapped magnetic flux [103,105], and (iii) weakening mechanical and thermal coupling at the Nb/Cu interface [9], all of which contribute to the Q-slope at high accelerating gradients.

Our results reveal a complex, multi-dimensional optimization landscape for HiPIMS processing of Nb-on-Cu SRF cavities. At high $E_i$ and $f_i$ (e.g., $f_i = 66\%$, $E_i = 200$ eV), intense ion bombardment produces well-adhered interfaces through deep ballistic mixing and smooth surfaces ($\sigma \approx 1.9$Å) via effective atomic peening. However, we find that these benefits come at significant cost: (i) excessive Cu enrichment at the interface (63 at.% Cu) generates high vacancy concentrations that degrade interfacial quality, (ii) upward Cu migration contaminates the

film surface, inhibiting complete crystallization and potentially compromising superconducting properties, (iii) substantial sputtering reduces deposition efficiency, and (iv) enhanced Ar incorporation introduces additional impurities.

Conversely, at low $E_i$ and $f_i$ (e.g., $f_i = 10\%$, $E_i = 50$ eV), insufficient ballistic mixing produces Cu-depleted interfaces (44 at.% Cu) that also retain excess vacancies (1.56 (40 at.% Cu) vs. 1.65 (60 at.% Cu) per 100 atoms for their composition), albeit slightly fewer than at high energies. More critically, we observe that limited adatom mobility and statistical roughening produce rough surfaces ($\sigma \approx 5.7$Å at $E_i = 50$ eV) that directly degrade RF performance, while weak interfacial mixing compromises mechanical adhesion.

The optimal processing regime lies at intermediate conditions (approximately $f_i = 33\%$, $E_i = 70$–$100$ eV), where the interfacial composition approaches $Cu_{45}Nb_{55}$ and vacancy concentration reaches a minimum (1.54 per 100 atoms). Under these conditions, sufficient ballistic mixing ensures good adhesion while avoiding excessive Cu enrichment, moderate atomic peening provides acceptable surface smoothness ($\sigma \approx$ 2Å) without extreme stress accumulation, and controlled crystallization kinetics minimize both interface defects and surface roughness.

Compared to prior MD studies of energetic Nb film growth that primarily characterized surface morphology and ion-bombardment effects [19,28–30], the present work specifically isolates the transient interfacial amorphization mechanism, governing mechanical adhesion, and quantifies the non-monotonic relationship between interfacial composition and vacancy defect density, the atomistic precursor to nanoscale void formation, thereby establishing the missing atomistic link to the empirically observed optimal bias window for SRF cavity performance. Consequently, this work provides direct atomistic guidance for optimizing the HiPIMS process to achieve superior SRF cavity fabrication. The identification of $Cu_{45}Nb_{55}$ as the optimal interfacial composition, corresponding to specific processing conditions ($f_i \approx 33\%$, $E_i \approx 70$–$100$ eV), offers a concrete, testable target for experimental process development. Rather than treating higher ion energy as universally beneficial, we conclude that process optimization should aim for controlled tuning of deposition parameters to target an interfacial composition near the identified optimum—a concrete, testable route for improving Nb-on-Cu SRF cavity performance.

## 5. Conclusion

In this work, we have employed comprehensive MD simulations to unravel the complex mechanisms governing the energetic deposition of Nb thin films on Cu substrates, to provide an atomistic roadmap for improving the SRF cavity performance. Our simulations demonstrate that high ion energy and ionization fraction lead to transient surface roughening. This is driven by the rapid crystallization of a metastable amorphous interface, whose composition is controlled by the deposition parameters.

Crucially, this atomistic finding provides a direct physical rationale for recent experimental observations in Nb-coated SRF cavities. Independent studies by Wang et al. [25] and Carlos et al. [26] demonstrated that cavity performance and critical current density reach a distinct optimum at an intermediate substrate bias of approximately −75 V, degrading at both higher and lower biases. Our simulation results are consistent with this empirical trend: the intermediate energy range of 70–100 eV (at $f_i \approx 33\%$) produces an interfacial composition near $Cu_{45}Nb_{55}$ that minimizes the retention of performance-degrading vacancy defects, while both lower and higher energies drive the composition away from this optimum and increase the defect density.

The main conclusion of this study is the identification of an optimal processing window for defect minimization. We revealed a non-monotonic, U-shaped relationship between the Cu concentration in the amorphous interface and the final density of vacancy defects. The minimum number of defects occurs at an optimal composition of ~45% Cu. This insight fundamentally shifts the optimization paradigm from simply maximizing interfacial mixing to precisely controlling the deposition parameters to achieve this ideal composition. Conditions that require too low-energy deposition fail to create sufficient mixing, while those conditions consisting of too high-energy surpass the optimum point, creating a Cu-rich interface that is also prone to higher defect formation.

These localized vacancy clusters represent the atomistic nucleation stage of nanoscale void formation and can degrade both interfacial adhesion and SRF cavity performance. While our parametric variation of $f_i$ captures the fundamental atomistic effects of different flux compositions, explicitly modeling the macroscopic temporal structure of the HiPIMS pulse, including duty cycle and dynamic ion-flux evolution, represents an important direction for future multiscale simulation efforts.

## CRediT authorship contribution statement

**Milad Ghaemi:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kostas Sarakinos:** Writing – review & editing, Validation, Investigation, Formal analysis. **Flyura Djurabekova:** Writing – review & editing, Validation, Supervision, Resources, Funding acquisition. **Alvaro Lopez-Cazalilla:** Writing – review & editing, Supervision, Methodology. **Guillaume Jonathan Rosaz:** Writing – review & editing, Resources, Funding acquisition. **Sergio Calatroni:** Writing – review & editing, Resources, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data for this article can be found online at doi:10.1016/j.mtcomm.2026.115373.

## Data availability

Data will be made available on request.

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